

6.3 Solve the equations

$$2x - y + z = 4$$

$$x + y + z = 3$$

$$3x - y - z = 1$$

Q: Write a function (find_prime_number) that finds if the input number to the function is a prime number or not. if the input number is a prime, it will display (P is a prime number) where P is the input number; otherwise; it will display (P is NOT a prime number)

8.8 Write a program to compute the sum of the series $1^2 + 2^2 + 3^2 \dots$ such that the sum is as large as possible without exceeding 1000. The program should display how many terms are used in the sum.

8.14 When a resistor (R), capacitor (C) and battery (V) are connected in series, a charge Q builds up on the capacitor according to the formula $Q(t) = CV (1 - e^{-t/RC})$ if there is no charge on the capacitor at time $t = 0$. The problem is to monitor the charge on the capacitor every 0.1 seconds in order to detect when it reaches a level of 8 units of charge, given that $V = 9$, $R = 4$, and $C = 1$. Write a program which displays the time and charge every 0.1 seconds until the charge first exceeds 8 units (i.e. the last charge displayed must exceed 8). Once you have done this, rewrite the program to display the charge only while it is strictly less than 8 units.